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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO4 ASSESSMENT TOOL 2

22.08.2026

3. Select any open education dataset containing student, school, examination, literacy, or enrollment information. Explore and analyze the data using grouping, comparison, correlation, outlier detection, and visualization techniques. Publish the analysis with appropriate charts and conclusions.

AIM: To explore and analyze an open education dataset using grouping, comparison, correlation analysis, outlier detection, and data visualization techniques to identify important patterns and draw meaningful conclusions about student educational performance.

TOOLS USED: Python,Pandas,NumPy,Matplotlib,Seaborn ,VS Code.

DATA USED: Education Dataset .

DATA GROUPING ANALYSIS : The data was grouped based on department to compare:

- Average internal marks
- Average assignment marks
- Average examination marks
- Average attendance
- Number of students

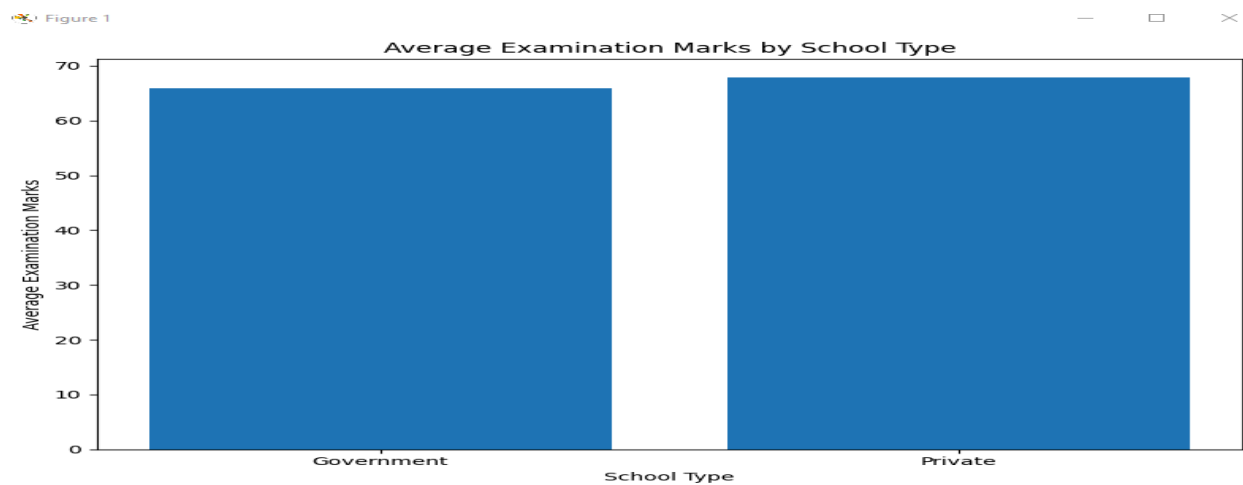
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===== DEPARTMENT ANALYSIS =====
  Department  Average_Internal_Marks  ...  Average_Attendance  Number_of_Students
0    CIVIL          34.663774  ...          78.288491          106
1     CSE          35.091154  ...          76.615192          104
2     ECE          34.611165  ...          77.259709          103
3     EEE          36.114808  ...          76.550769          104
4     MECH          34.626747  ...          75.517590           83

[5 rows x 6 columns]

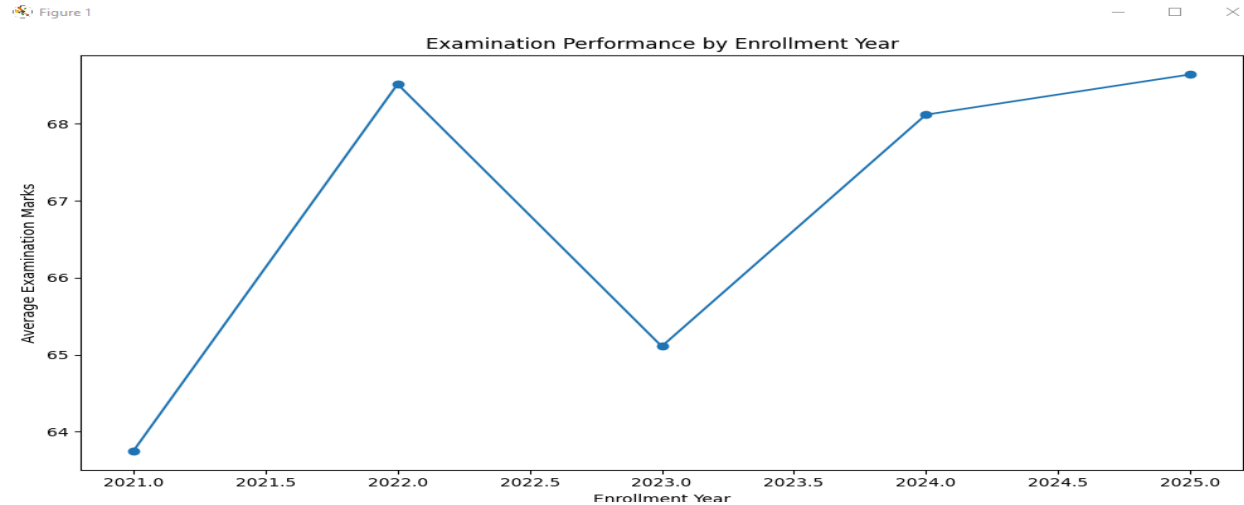
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COMPARISON ANALYSIS: Students were compared based on school type, including: Government, Private. Average examination marks, attendance, and study hours were compared between the two school types.



ENROLLMENT YEAR ANALYSIS:

Student performance was analyzed according to enrollment year. The average examination marks, attendance, study hours, and number of students were compared across different enrollment years.

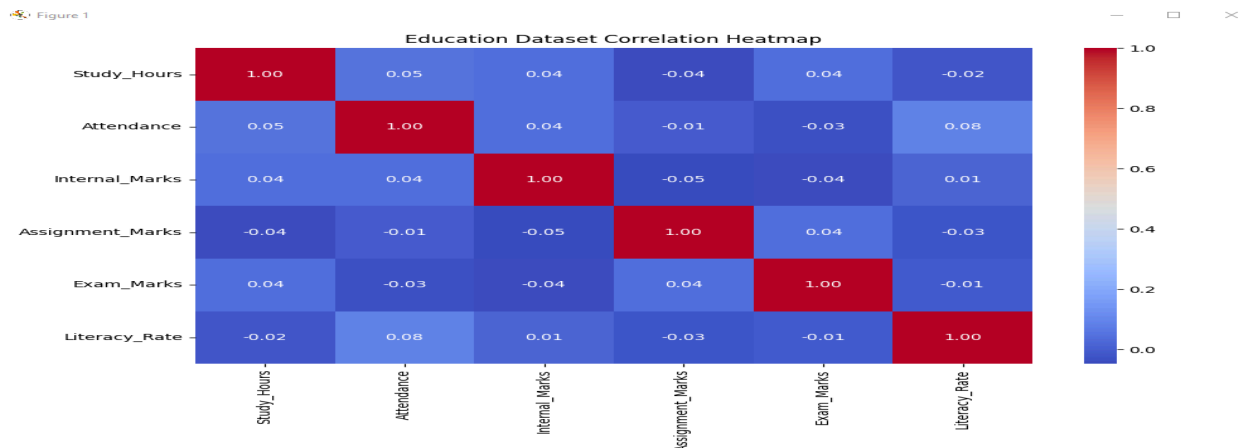


CORRELATION ANALYSIS:

Correlation analysis was performed to identify relationships between:

- Study Hours
- Attendance
- Internal Marks
- Assignment Marks
- Examination Marks
- Literacy Rate

The correlation coefficient ranges from **-1 to +1**. A positive value indicates a positive relationship, while a negative value indicates an inverse relationship.



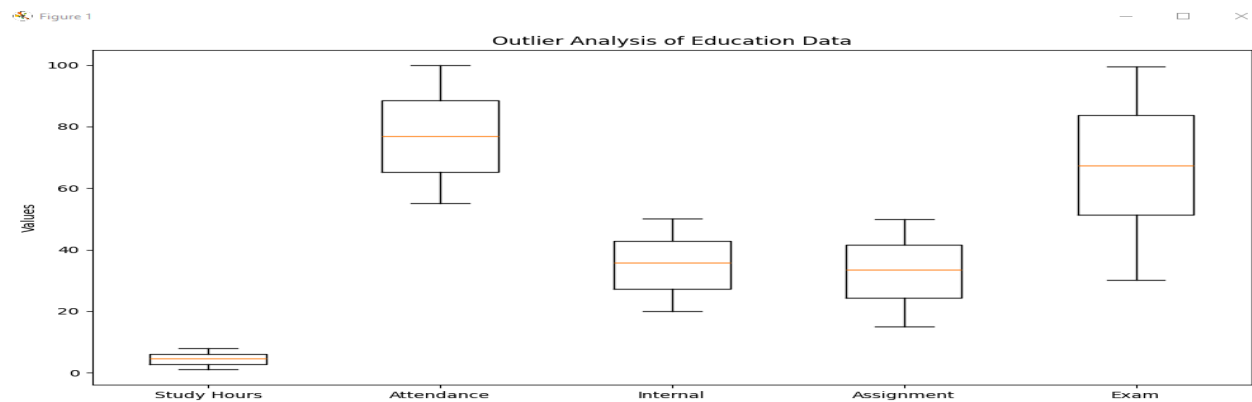
OUTLIER DETECTION:

The **Interquartile Range (IQR) method** was used to identify unusual observations in:

- Study Hours
- Attendance
- Examination Marks

The IQR method uses the first quartile (Q1), third quartile (Q3), and the following formula:

$IQR = Q3 - Q1$. Values outside the lower and upper limits were considered potential outliers.



KEY FINDINGS:

1. Best Performing Department

The ECE department has the highest average examination marks, with an average score of 68.56.

2. School Type Comparison

Private school students have the highest average examination marks, with an average score of 67.81, compared with Government school students.

3. Study Hours and Examination Performance

The correlation between study hours and examination marks is 0.04. This indicates a very weak positive relationship between the two variables in the analyzed dataset.

4. Attendance and Examination Performance

The correlation between attendance and examination marks is -0.03. This indicates a very weak negative relationship between attendance and examination marks in this dataset.

5. Best Enrollment Year

Students enrolled in 2025 have the highest average examination marks, with an average score of 68.64.

CONCLUSION:

The education dataset was successfully analyzed using Python, Pandas, NumPy, Matplotlib, and Seaborn. Grouping and comparison techniques were used to examine academic performance across departments, school types, and enrollment years.

The analysis found that ECE had the highest average examination marks at 68.56, while Private school students had the highest average at 67.81. Students enrolled in 2025 recorded the highest average examination marks at 68.64.

The correlation between study hours and examination marks was 0.04, while the correlation between attendance and examination marks was -0.03. Both relationships were very weak in this dataset.

Overall, the analysis demonstrates how data exploration, statistical analysis, correlation, outlier detection, and visualization can be used to identify educational patterns and support data-driven decision-making.

7. Select any open agriculture dataset containing crop production, rainfall, soil, irrigation, fertilizer, or agricultural productivity data. Analyze relationships between variables, identify outliers, create meaningful groups, study trends over time, and visualize the results using charts and maps.

AIM: To analyze an open agriculture dataset containing crop production, rainfall, soil/agricultural inputs, fertilizer, irrigation-related factors, and agricultural productivity data by studying relationships between variables, identifying outliers, creating meaningful groups, analyzing trends over time, and visualizing the results using charts and maps.

DATASETS USED: Crop Production Dataset .

TOOL USED: Python,Pandas,NumPy,Matplotlib,Seaborn ,VS Code.

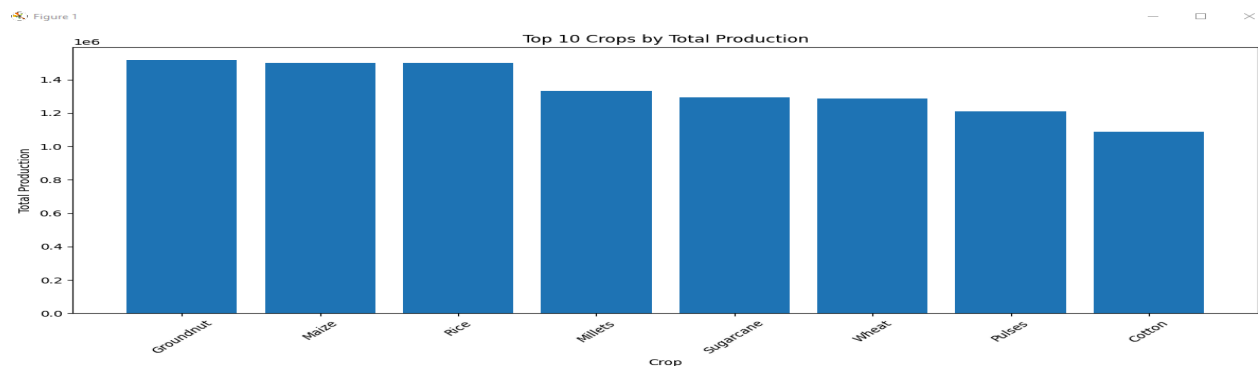
CROP GROUPING ANALYSIS:

The agriculture data was grouped according to **crop type**.

For each crop, the following were analyzed:

- Total production
- Average yield
- Total cultivated area

This grouping helped identify the crops with higher production and productivity.



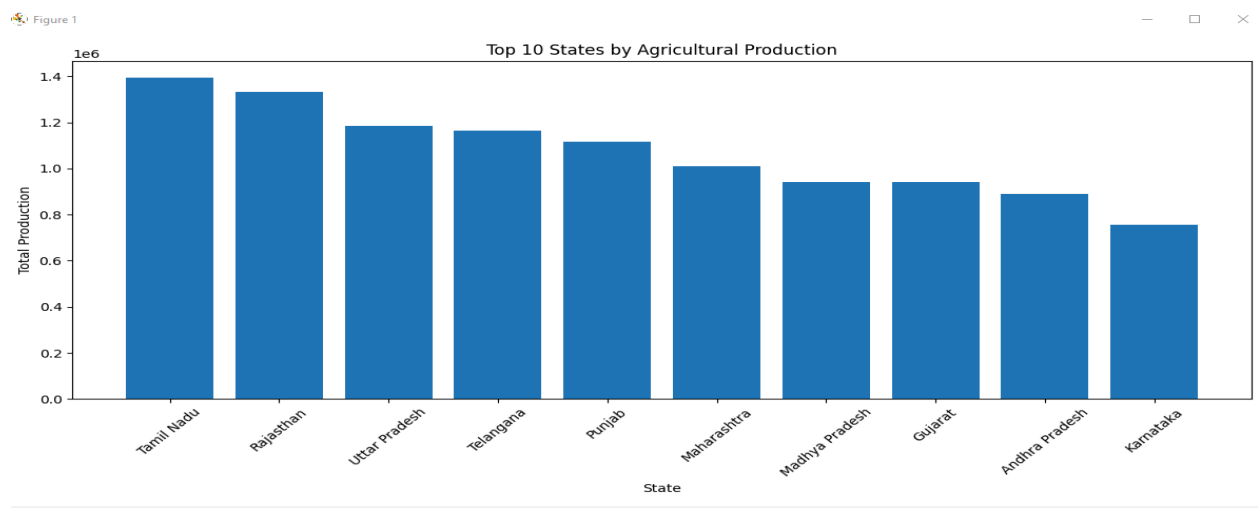
STATE GROUPING ANALYSIS:

The data was grouped according to **state** to compare agricultural performance across different regions.

The following parameters were analyzed:

- Total production
- Average yield
- Total cultivated area
- Average annual rainfall

This helped identify the states with higher agricultural production.



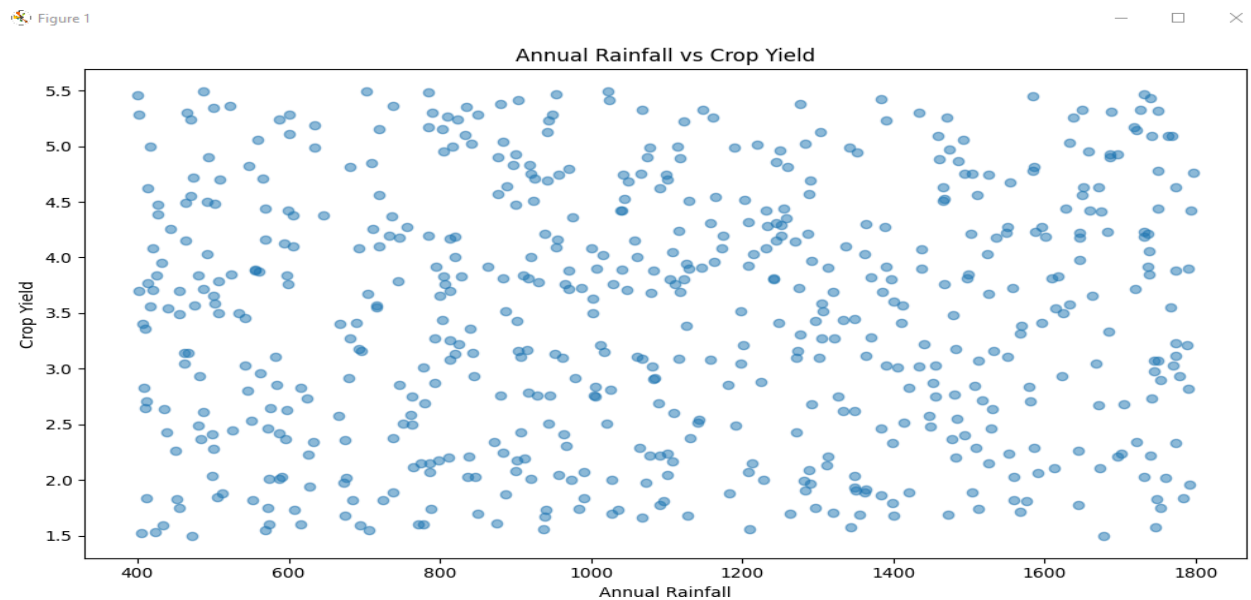
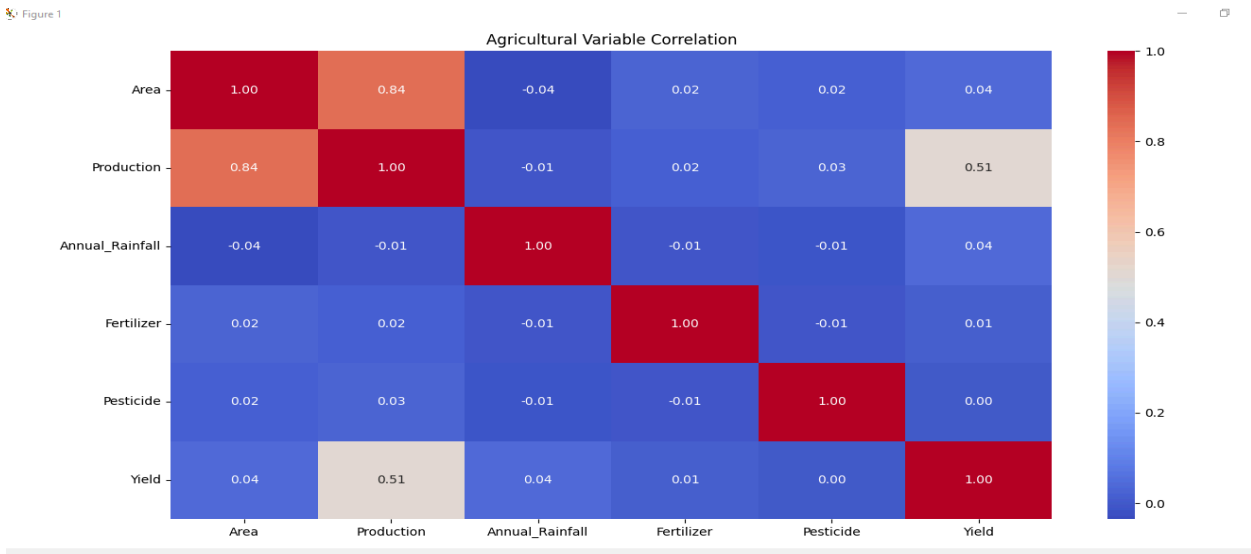
RELATIONSHIP AND CORRELATION ANALYSIS:

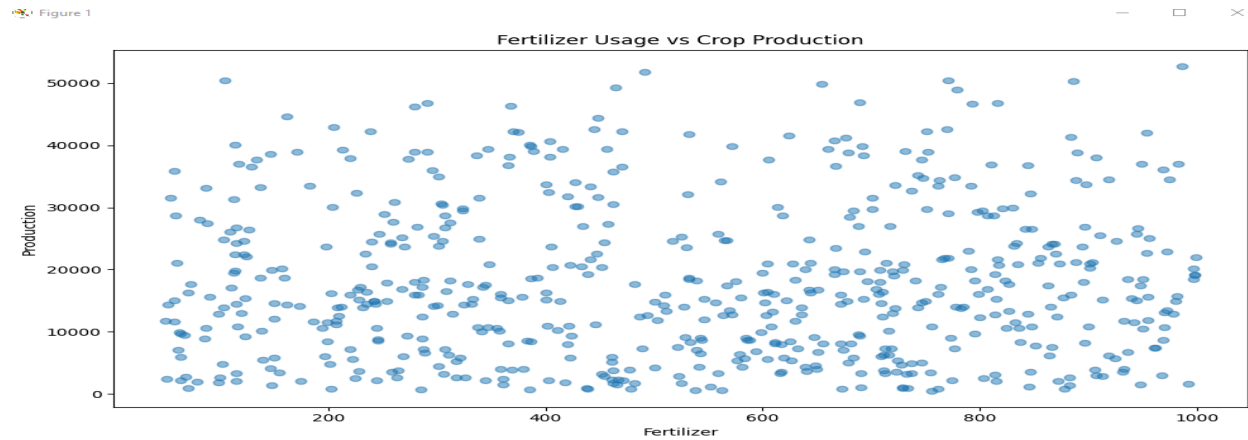
Correlation analysis was performed to study relationships between:

- Area
- Production
- Annual rainfall
- Fertilizer
- Pesticide
- Yield

The correlation coefficient was used to determine whether the variables had positive or negative relationships.

A value close to **+1** indicates a strong positive relationship, while a value close to **-1** indicates a strong negative relationship.





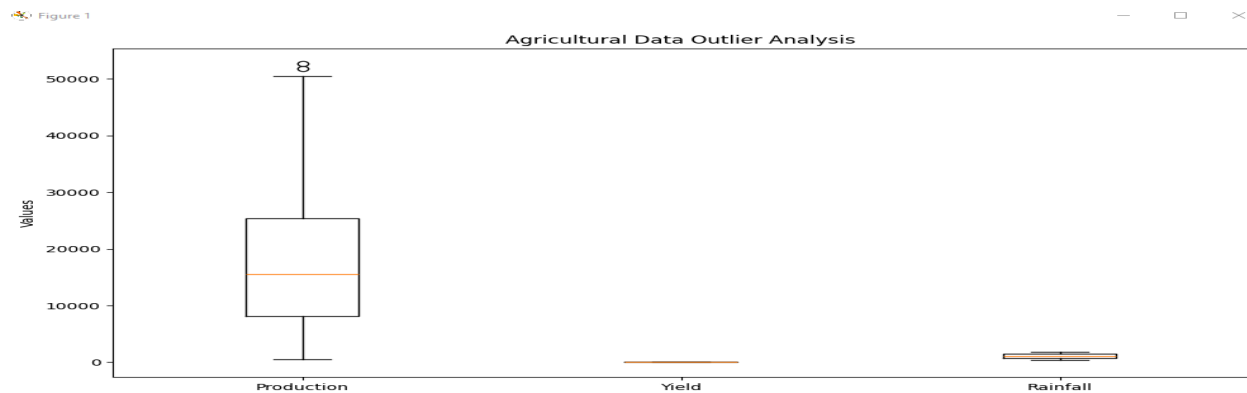
OUTLIER DETECTION:

The **Interquartile Range (IQR) method** was used to identify potential outliers in:

- Production
- Yield
- Annual Rainfall

The IQR was calculated using: $IQR = Q3 - Q1$

Values below the lower limit or above the upper limit were considered potential outliers.



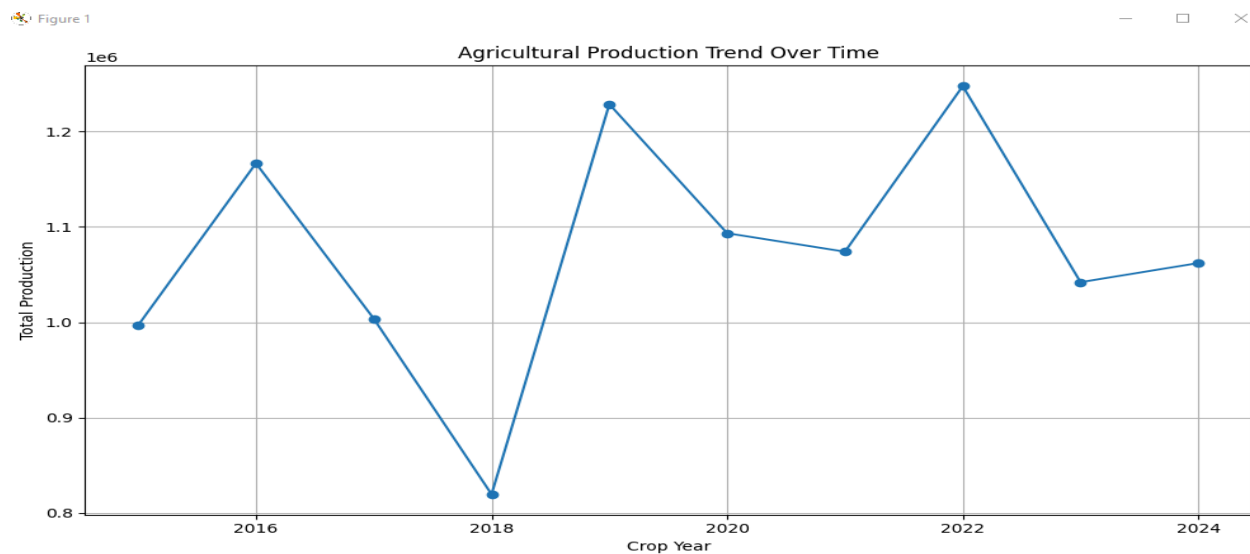
TIME-BASED TREND ANALYSIS:

Agricultural production was analyzed across different **crop years**.

The analysis compared:

- Total production
- Average yield
- Average annual rainfall over the years.

This helped identify changes and trends in agricultural production over time.



KEY FINDINGS:

1. RAINFALL AND YIELD

The correlation between rainfall and crop yield is 0.04.

2. FERTILIZER AND PRODUCTION

The correlation between fertilizer usage and production is 0.02.

3. BEST PRODUCTION YEAR

The year 2022 recorded the highest total agricultural production of 1247400.56.

CONCLUSION:

The agriculture dataset was successfully explored and analyzed using Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly. Grouping techniques were used to compare crops and states, while correlation analysis was used to study relationships between agricultural variables such as rainfall, fertilizer, production, area, and yield.

The IQR method was used to identify potential outliers, and time-based analysis was performed to understand changes in agricultural production over the years. Various charts were created to visualize crop production, state-wise production, relationships between variables, and agricultural trends. A geographical map was also used to represent agricultural production across states.

Overall, the analysis demonstrates how data analytics and visualization can be used to understand agricultural productivity, identify important patterns, detect unusual observations, and support data-driven agricultural planning and decision-making.

